AI in protein biology

Embeddings and structure prediction

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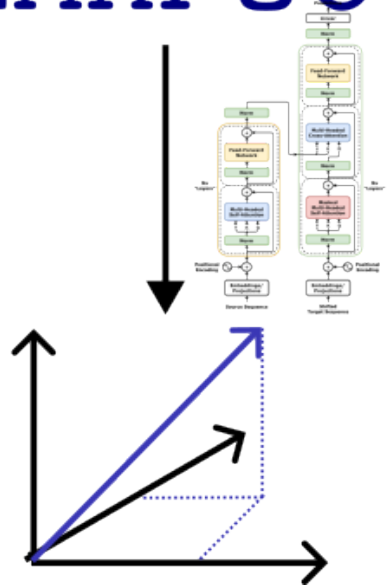
Outline

- Embeddings: vector representation of protein (sequences)
- Structure prediction: principle and model
- Strucural similarity search

Language models for proteic sequences

Model providing a latent representation of the sequences such that homologous sequences are close in latent space.

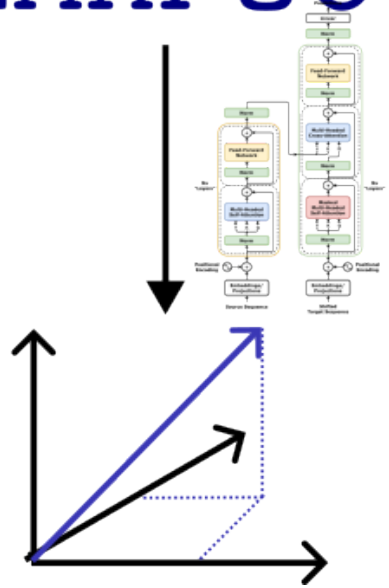
MFILMKPOCVSPK



Language models for proteic sequences

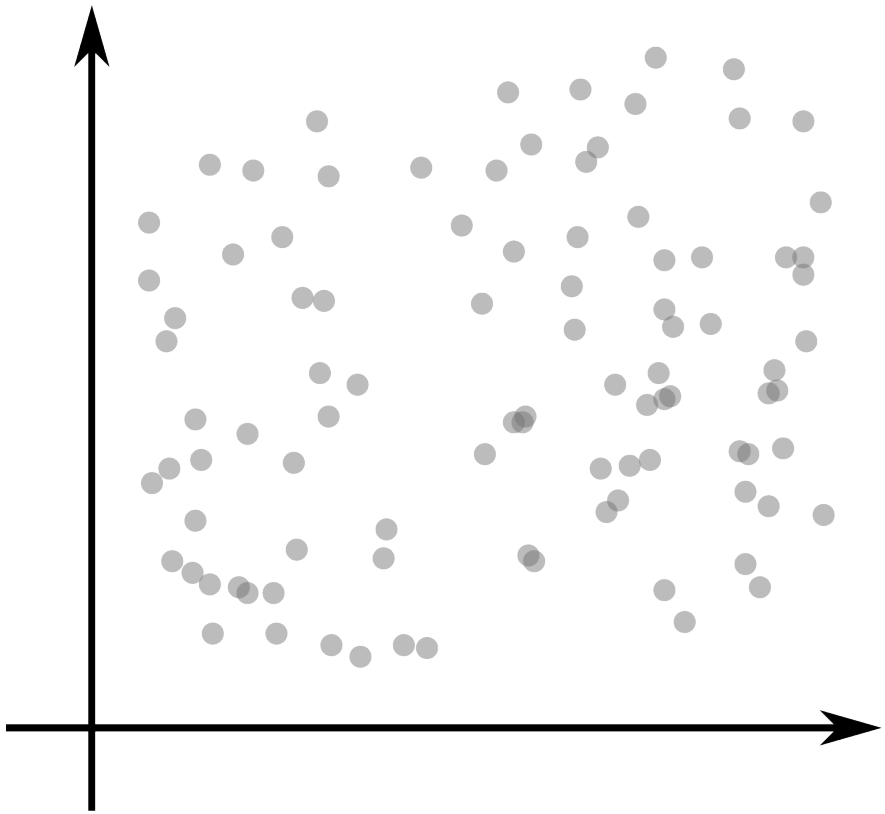
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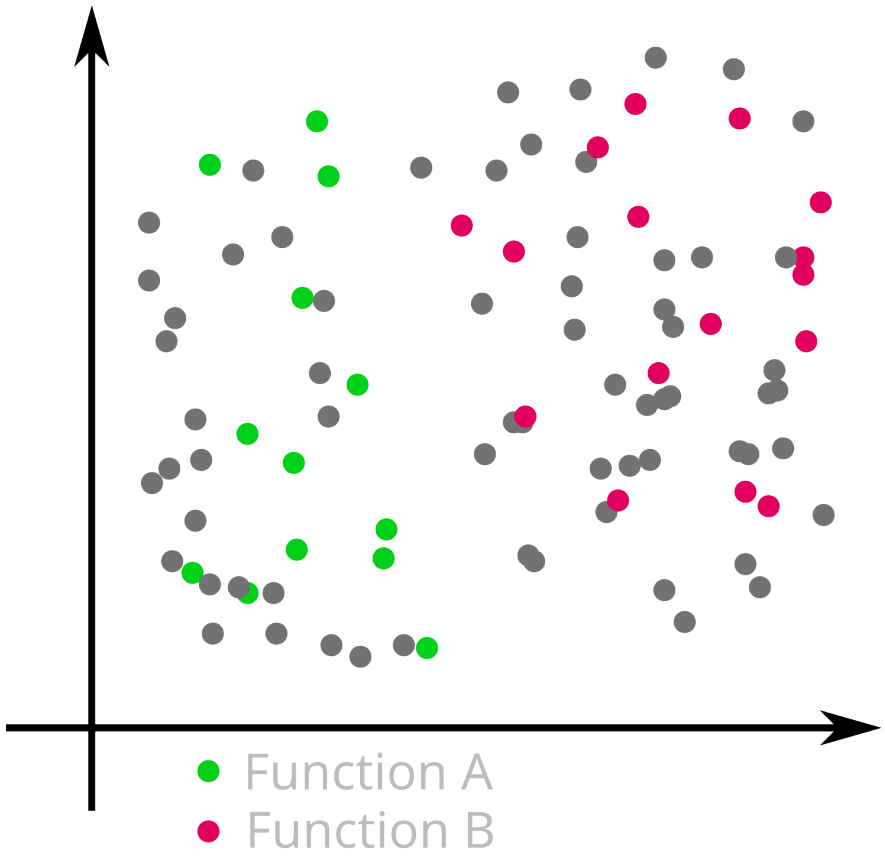


Neural network with Transformer architecture trained self-supervisedly on billion of proteic sequences.

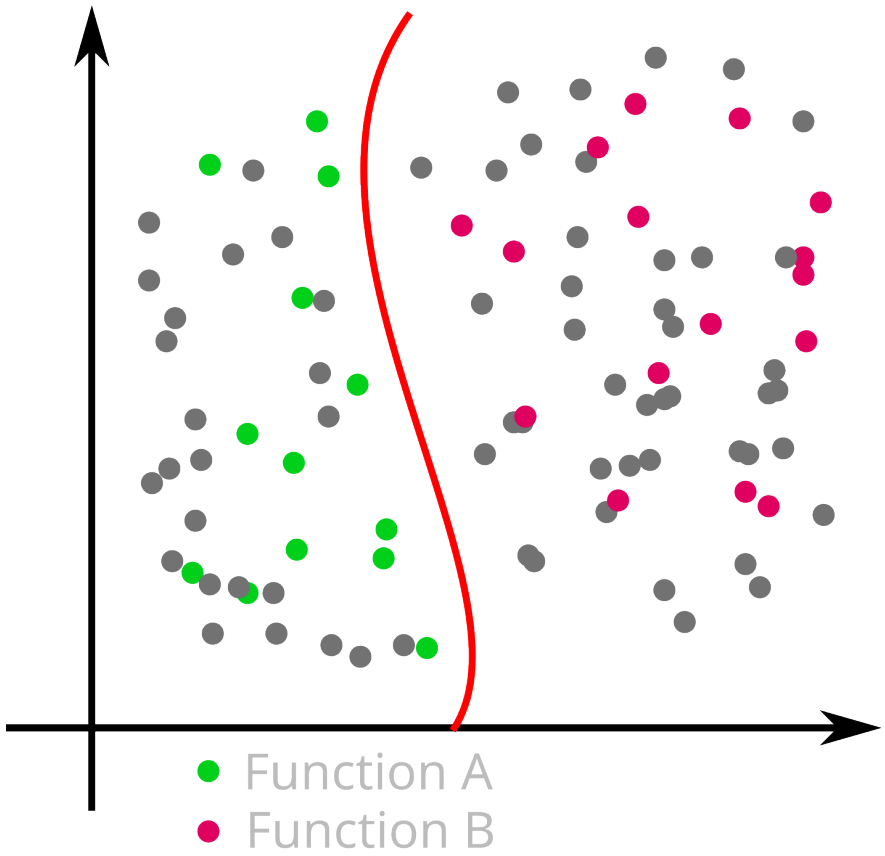
Protein embedding for function prediction



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Protein embedding for function prediction



Training of language models for proteic sequences

MFILMKPOCVSPK

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Training of language models for proteic sequences





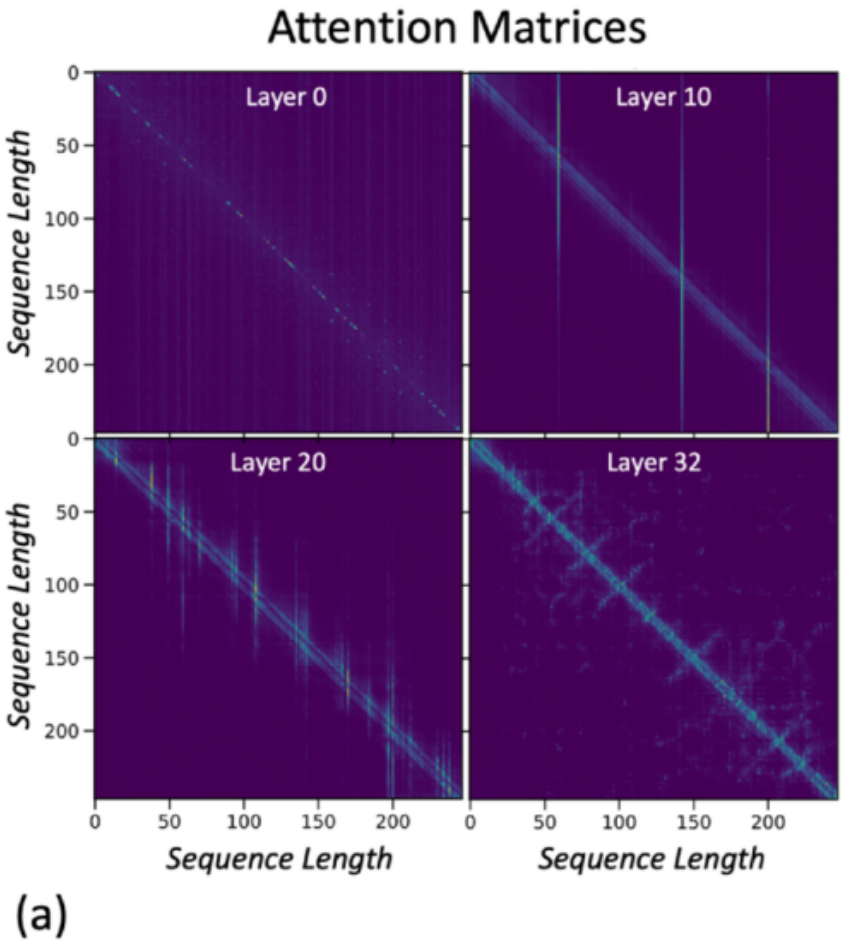
Training of language models for proteic sequences





To best predict the masked residues, the model is enforced to *pay attention to* the influencing residues.

Example of attention vs contact (P07477)





[Nayar et al. 25 PLOS Comp Biol]

Language Models as representation spaces

- barcodeBERT
- protT5
- ESM
- hyenaDNA
- ...

Attention learning

Text-to-text models

$$\text{Masked sequence} \xrightarrow{f_w} \text{Full sequence}$$

Text-to-text models

$$\text{Input sequence} \xrightarrow{f_w} \text{Output sequence}$$

Text-to-text models

$$\text{Input sequence} \xrightarrow{f_w} \text{Output sequence}$$

Goal

Learn w such that $f_w(x_i) \approx y_i$ on some training dataset $\{(x_i, y_i) | i = 1...n\}$

Introductory example: NL translation

Goal

Design an AI system that is able to translate texts from one natural language to another, *e.g.* English to French.

"Naïve" possibility:

- Use a pre-defined dictionary
- Couple the dictionnary with Formal Grammars for English (parsing) and French (Generative)

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Issue

A lot of language engineering involved, need to re-do the work for all (pairs of) languages.

Machine learning for translation (2010's)

Goal

Use a corpus of texts i in two versions each (e_i, f_i), in English and in French, respectively.

Attempt 1 for ML-based translation

Any guess?

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Word-by-word translation: learn a dictionnary.

$$p(y_i|x_i) \tag{1}$$

Brain equivalent

- Look at a word, and translate it
- Forget about this word

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Issues

- Lack of context "What a nice example" $\rightarrow$ "Quoi un gentil exemple"
- Words are not aligned (y_i does not corresponds to x_i, nor even same length: "A writing machine" $\leftrightarrow$ "Une machine à écrire"

Attempt 2

Store some information about the context, possibly both from the input and already preocessed output sequence.

- $p(h_i|x_1, ...x_i)$: h_i describes the state of the ith input word
- $p(s_i|y_1, ...y_{i-1})$: s_i describes the state of the ith output word
- $p(y_i|x_i, h_i, s_i)$

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- $p(y_i|x_i, h_i, s_i)$: the ith output should depend on the ith input and states

Issues

- Across languages, words does not align one-to-one

Attempt 3

Translate chunk (i.e. sentence) by chunk: learn a representation of the sentence, and then translate it.

[1] Note that vanilla RNNs have a vanishing gradient issue, prefer to use LSTM as in [Sutskever, NIPS'14]

Attempt 3

Translate chunk (i.e. sentence) by chunk: learn a representation of the sentence, and then translate it. Can be done using a encoder-decoder RNN[1] structure:

- $p(h_i|h_{i-1}, x_i)$
- $p(s_j|s_{j-1}, y_{j-1})$
- $p(y_j|h_L, s_j)$

Where L is the input sequence length.

[1] Note that vanilla RNNs have a vanishing gradient issue, prefer to use LSTM as in [Sutskever, NIPS'14]

Attempt 3: still some issues...

Brain equivalent

- Read a sentence, memorize its meaning
- Translate it

Any drawback?

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Pros and cons

- + Circumvented the alignment problem
- - Need to get very expressive representation to store all information of a sentence into a latent variable $\Rightarrow$ for long sentences, the information is distored

Translation challenge

Look at this dataset ($n = 1$) of translations:
Hello World!/enihcam wen eht gniyrt m'I/It works/sretirw rof tluciffid kool yam ti tuB
Hello World!/I'm trying the new machine/It works/But it may look difficult for writers

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Exercise

How would you translate "Hey/ti tog uoy kniht I/Well done!/!stargnoC"

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Exercise

How would you translate "Hey/ti tog uoy kniht I/Well done!/!stargnoC"
"Hey/I think you got it/Well done!/Congrats!"

How can a machine learn from such examples how to translate new texts?
What's your brain doing for translating this?

Translation challenge: needed structure

- h_i: input state (parity of current line number)
- s_j: output state
- $\alpha_{i,j}$: alignment score matrix between the ith input state and the jth output state.

By having the α at hand, you know what letter of the input data you have to translate.

What would be perfect

- $p(h_i|x_1, ...x_i)$
- $p(s_j|y_1, ...y_{j-1})$
- $p(y_j|h_{\alpha(j)}, x_{\alpha(j)}, s_j)$

Where $\alpha(j)$ is the word of the input sequence aligned to j.

Breakthrough

In 2015, [Bahdanau, Cho and Bengio] made a big breakthrough by learning s, h and α from the data!

Transformers

Transformer is an implementation of attention: it learns where to let flow the information in the network.

Self-suervised learning (masking)



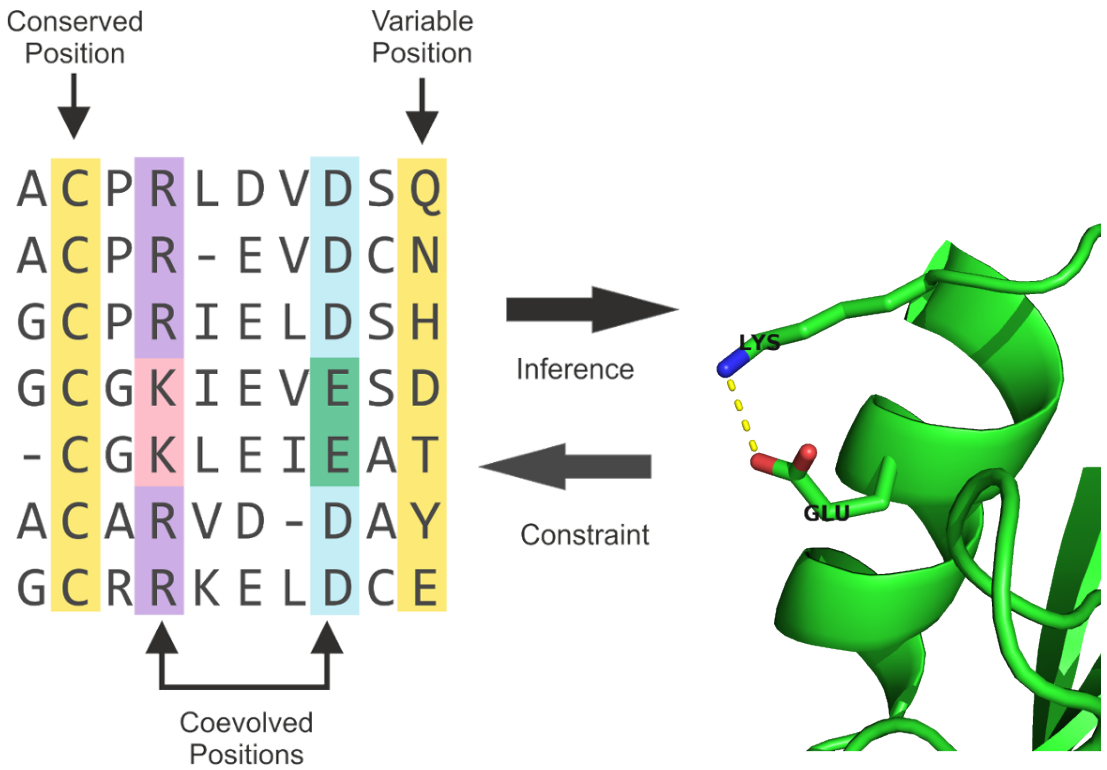


Best option: `https://ai-protein-biology.apps.math.cnrs.fr`
Alternative option: `https://tinyurl.com/ai-protein-biology`

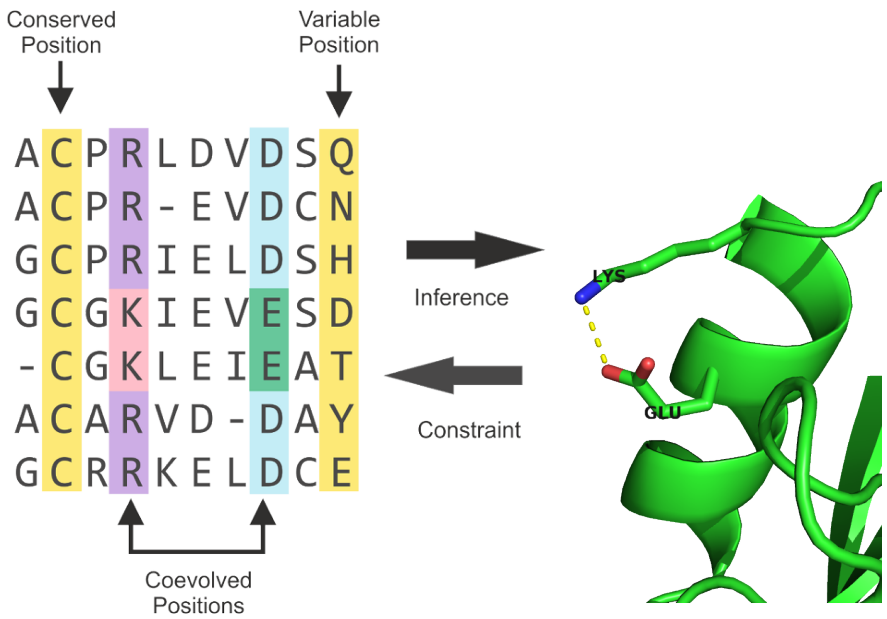
Structure prediction

Preserving the function: coevolution of residues

As protein function is vital, **evolution selects mutations preserving structures**.
Leading to **compensatory** mutations:



A simple approach for protein structure prediction



- Build or get multiple manio acid sequence alignments (**e.g.** in Pfam database)
- Quantify coevolution between positions in the sequence
- Infer what are the position in contact

What measure for co-evolution?

Conservation vs. co-evolution

A standard approach is to measure it through Mutual Information:

$$MI(i,j) = \sum_{a,b} p(x_i = a, x_j = b) \log \frac{p(x_i{=}a, x_j{=}b)}{p(x_i{=}a)\, p(x_j{=}b)}$$

Where

- x_i is the amino acid at position i
- $p(x_i = a)$ is estimated in the MSA by $\frac{\#\text{sequences having "a" at position } i}{N}$
- N the number of sequences in the MSA
- $p(x_i = a, x_j = b)$ is estimated in the MSA by $\frac{\#\text{sequence having "a" at } i \text{ and "b" at } j}{N}$

In paractice you need $N > 1,000$ to have reasonable estimation of $p(x_i = a, x_j = b)$.

From heuristic to learning

Drawbacks:

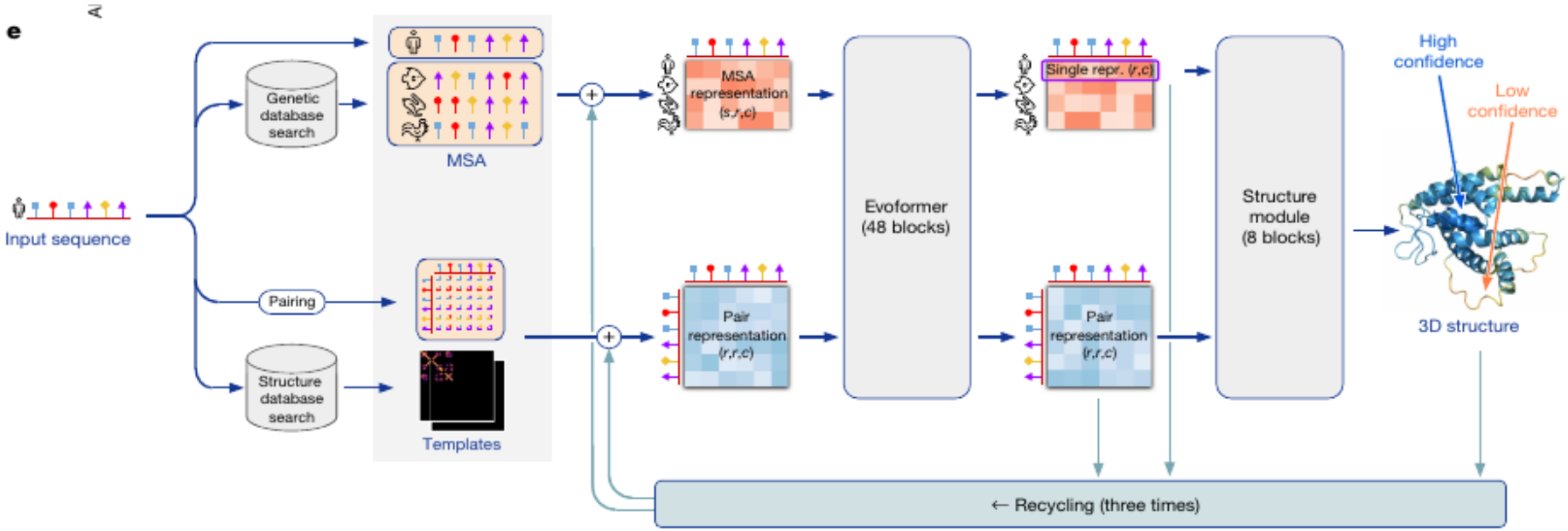
From heuristic to learning

Drawbacks:

- Spurious correlation/noisy estimate of MI when having few aligned sequences
- Indirect associations
- MI may not be the best option to detect correlated mutation due to 3D interaction

This is where Transformers have been used: localize portion in the MSA that are of interest to predict 3D interactions.

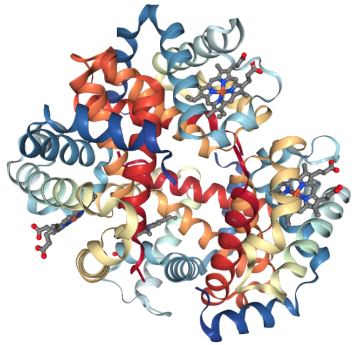
Alphafold2 model



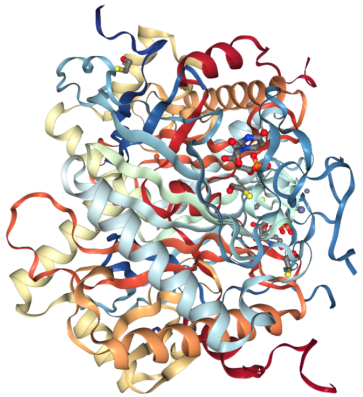
Protein structure prediction as "language translation"

Task: predict the structure from sequence

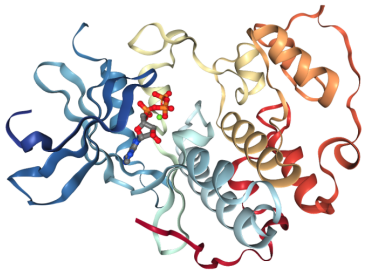
```
>1A3N:A|PDBID|CHAIN|SEQUENCE
VLSPADKTNVKAAWGKVGAHAGEYGAEALER
MFLSFPTTKTYFPHFDLSHGSAQVKGHGKKV
ADALTNAVAHVDDMPNALSALSDLHAHKLRV
DPVNFKLLSHCLLVTLAAHLPAEFTPAVHAS
LDKFLASVSTVLTSKYR
```



```
>1HXP:A|PDBID|CHAIN|SEQUENCE
MTQFNPVDHPHRRYNPLTGQWILVSPHRAKRPW
EGAQETPAKQVLPAHDPDCFLCAGNVRVTGDKN
PDYTGTYVFTNDFAALMSDTPDAPESHDPLMRC
QSARGTSRVICFSPDHSKTLPELSVAALTEIVK
TWQEQTAELGKTYPWVQVFENKGAAMGCSNPHP
HGQIWANSFLPNEAEREDRLQKEYFAEQKSPML
VDYVQRELADGSRTVVETEHWLAVVPYWAAWPF
ETLLLPKAHVLRITDLTDAQRSDLALALKKLTS
RYDNLFQCSFPYSMGWHGAPFNGEENQHWQLHA
HFYPPLLRSATVRKFMVGYEMLAETQRDLTAEQ
AAERLRAVSDIHFRESGV
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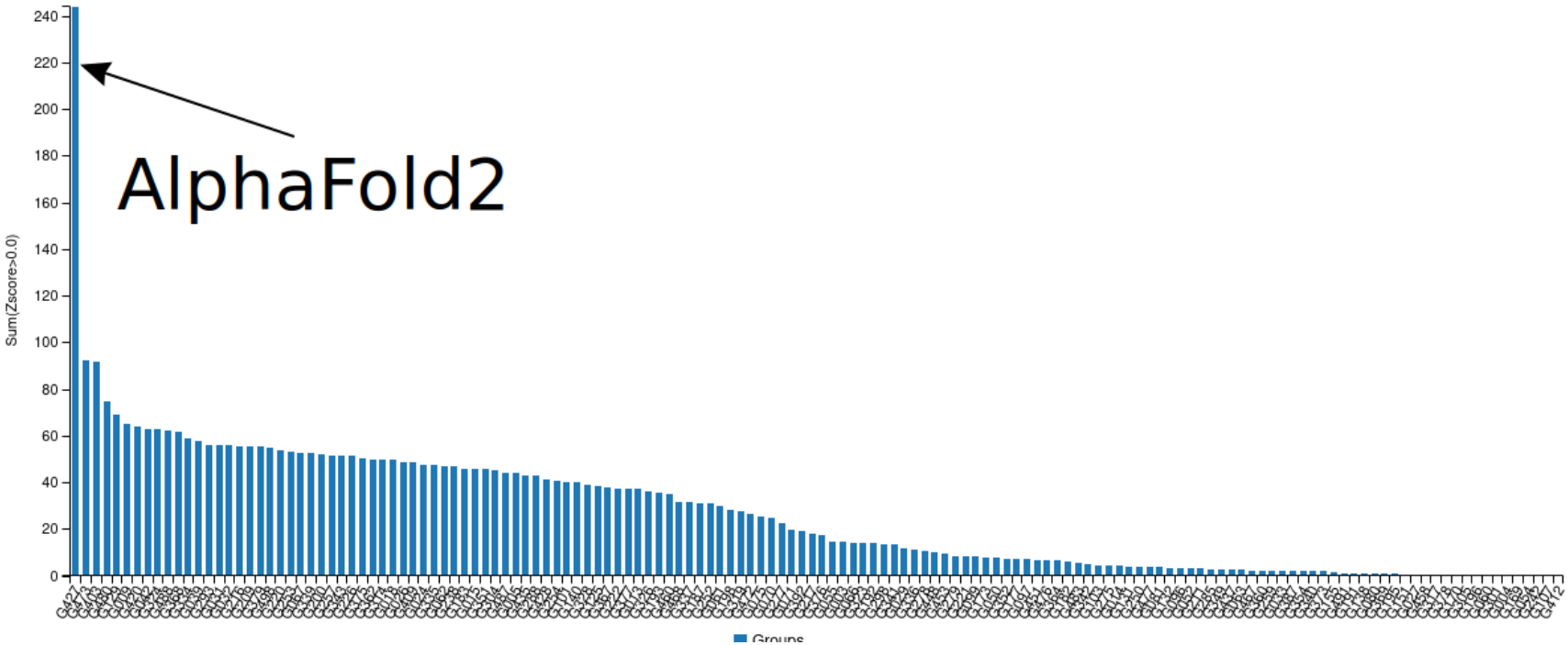


```
>1HCK:A|PDBID|CHAIN|SEQUENCE
MENFQKVEKIGEGTYGVVYKARNKLTGEVVAL
KKIRLDTETEGVPSTAIREISLLKELNHPNIV
KLLDVIHTENKLYLVFEFLHQDLKKFMDASAL
TGIPLPLIKSYLFQLLQGLAFCHSHRVLHRDL
KPQNLLINTEGAIKLADFGLARAFGVPVRTYT
HEVVTLWYRAPEILLGCKYYSTAVDIWSLGCI
FAEMVTRRALFPGDSEIDQLFRIFRTLGTPDE
VVWPGVTSMPDYKPSFPKWARQDFSKVVPPLD
EDGRSLLSQMLHYDPNKRISAKAALAHPFFQD
VTKPVPHLRL
```



CASP14 (2020)

"The big leap forward"



Alphafold2: attention-based learning on protein sequence alignments [Casp14.]
Nature's article.

Modern structural similarity search

Rise of new data: predicted structures

- AlphaFoldDB >200 millions of structures
- ESMAtlas >750 millions of structures

Open new avenues for protein function identification, study of evolutionary relations, etc.

The need for efficient structure comparison

Before 2020 structure comparison tools (Dali, TM-Align) were suitable for small structure databases.
Usual algorithms are non-local, preventing to use efficient one-dimensional dynamic programming.

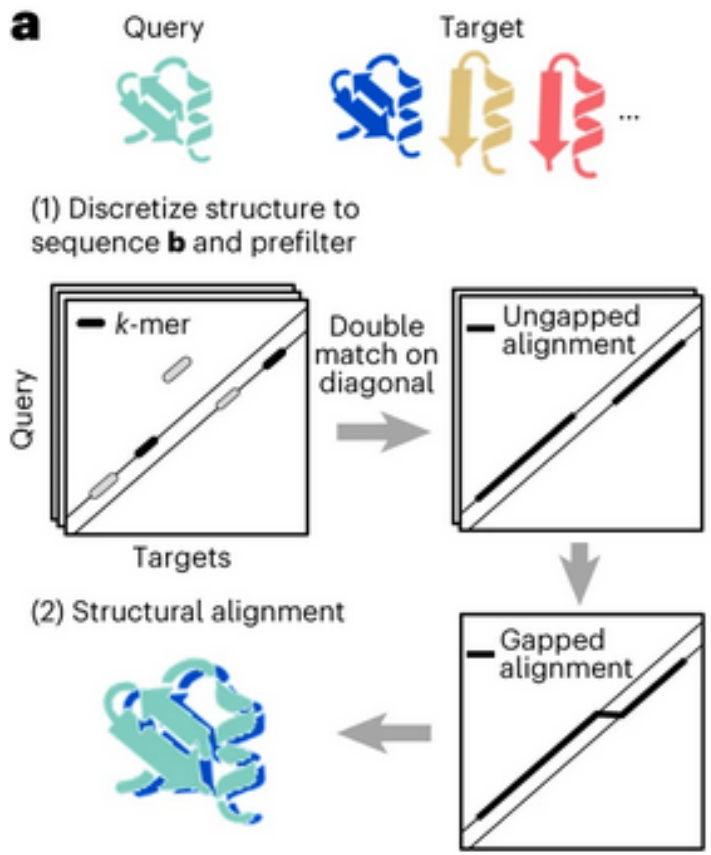
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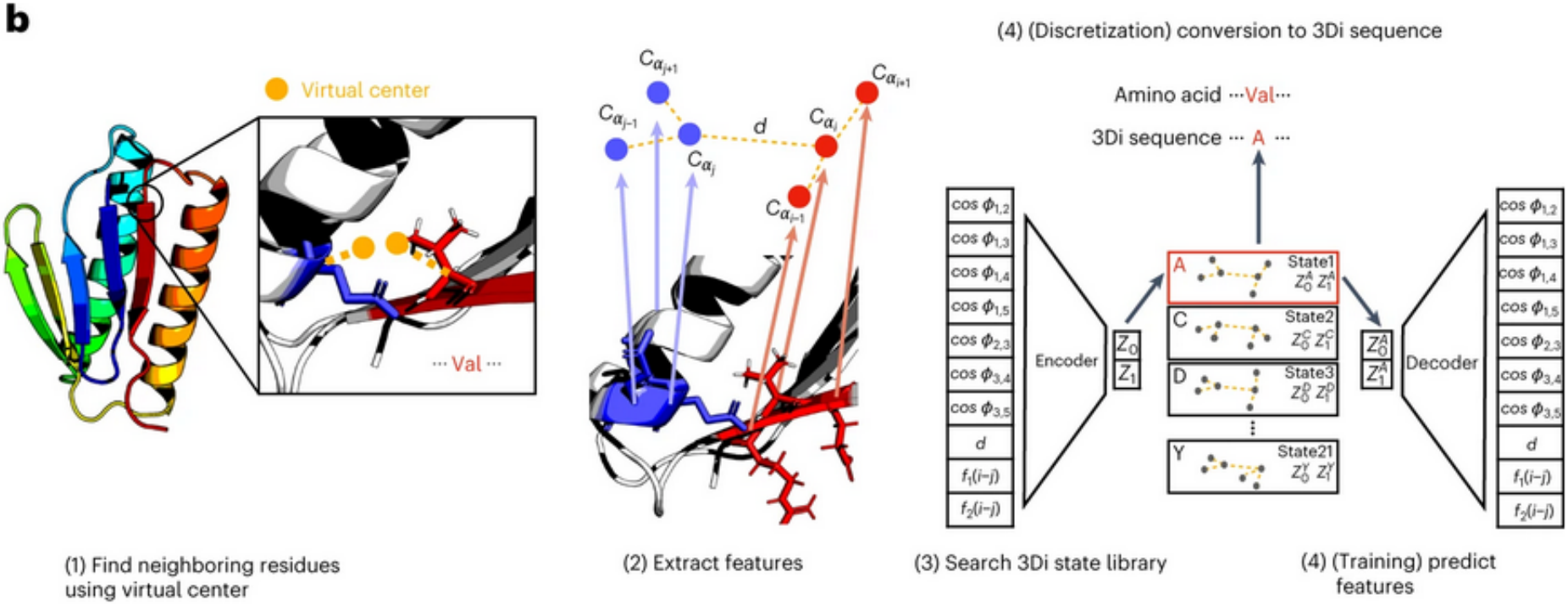
Foldseek

Foldseek discretizes the structure into a sequence over a 20-letter alphabet (3Di), and uses previously developed sequence-sequence search tool.

MMseqs/FoldSeek search principle



3Di


b
Virtual center
··· Val ···
(1) Find neighboring residues using virtual center
d
(2) Extract features
(4) (Discretization) conversion to 3Di sequence
Amino acid ···Val···
3Di sequence ··· A ···
Encoder
Decoder
State1
State2
State3
State21
(3) Search 3Di state library
(4) (Training) predict features